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Content-Centric Modelling of Rage-Bait Lifecycles on Social Media with Toxicity Feedback

An IVFS Compartmental ODE Model
Calibrated on the Higgs Twitter Dataset

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Abstract

We develop a deterministic compartmental ordinary differential equation model for the lifecycle of rage bait content on social media. The model tracks four content states (ignored, viral, fatigued, and suppressed) together with a latent toxicity variable τ and an active user pool subject to toxicity dependent attrition. Throughout this report, toxicity means the toxicity of rage baiting: a reply driven, high arousal discussion pressure represented by τ and by the feedback loop it creates with viral spread and user loss. It is not a direct text classifier score. We calibrate the spread block on the SNAP Higgs Twitter retweet stream, use same dataset reply activity as the observable proxy for this toxicity, and compare its scale against an external Jigsaw text toxicity reference. The fitted model produces a clear amplification threshold at $\alpha_{\text{crit}} \approx 0.3165$, below which viral spread is not self sustaining. Above this threshold, stronger amplification raises long run viral volume and toxicity while lowering user retention. A core feature of the model is the toxicity feedback loop: viral content drives reply activity and latent toxicity, which amplifies further spread and accelerates user departure. Our main contribution is the content centric framing: unlike most existing compartmental models for social media dynamics, which classify users into behavioural states, we classify content items into lifecycle compartments and couple them to a toxicity state and user attrition. The principal limitation is that the toxicity block is proxy grounded rather than directly calibrated to same platform text toxicity, so the pressure parameters are only partially identified from the available data.

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1 Background

1.1 Algorithmic amplification and platform health

Social media platforms use recommendation algorithms to select and rank content for users. These algorithms are typically optimized for engagement metrics such as clicks, shares, and time-on-platform. Recent work shows that engagement-optimized algorithms tend to amplify politically divisive, morally charged, and emotionally provocative content [5, 6, 4]. Huszár et al. [5] conducted a large-scale audit of Twitter's algorithmic timeline and found that the algorithm amplified content from political parties, with particularly strong effects for politically right-leaning sources in several countries. Milli et al. [6] showed that content promoted by engagement-based algorithms tends to be more divisive and less satisfying to users than content they would have chosen themselves. Brady et al. [4] demonstrated that moral-emotional language in social media posts significantly increases their diffusion, with each additional moral-emotional word raising the retweet rate by roughly 20%.

This suggests a tradeoff at the platform level. Amplification can increase short-run engagement, but it may simultaneously intensify harmful discussion dynamics, increase user fatigue, and erode long-run platform health. Beknazar-Yuzbashev et al. [7] give experimental evidence for this: a field experiment showed that exposure to toxic content on social media reduced subsequent user engagement, suggesting that toxicity imposes real costs on platform participation.

1.2 Compartmental models for information spread

Compartmental models from epidemiology have been applied widely to information dynamics. The classical SIR (Susceptible-Infected-Recovered) model partitions a population into compartments and describes transitions between them using ordinary differential equations. Daley and Kendall [3] noted the analogy between disease transmission and information spread early on and proposed a rumour-spreading model with similar compartmental structure. In the social media setting, De Domenico et al. [2] used the Higgs Twitter dataset to study the anatomy of a scientific rumour, fitting epidemic-style models to retweet cascades.

The DABAR modelling procedure taught in this course (*Define, Assume, Build, Assess, Report*) is the framework we follow. We define the modelling question around platform amplification tradeoffs, state our assumptions about aggregate content dynamics, build a six-state compartmental ODE system, assess it against empirical data and analytical checks, and report the results with clear attention to limitations.

The IVFS model we develop falls into the category of a *deterministic dynamical model* in the language of the course. It is a system of first-order autonomous ODEs whose state variables are continuous and whose parameters are either calibrated from data or fixed at scenario values. The model is not stochastic, not optimization-based, and not probabilistic. Its core analytical tools are equilibrium analysis, linearized stability, threshold quantities, and sensitivity indices, all of which are covered extensively in the course material.

1.3 The content-centric gap

Most existing compartmental models for social-media dynamics classify *users* into behavioural states. For example, an SVFR-type model might place users into susceptible, viewing, forwarding, and recovered compartments [8, 9]. Toxicity-focused models similarly track user infection states, where "infection" represents adoption of toxic behaviour [10, 11]. These models are valuable, but they require assumptions about individual user decision-making and local network structure that are difficult to validate at platform scale.

Our model takes a different approach. Instead of compartmentalizing users, we compartmentalize *content items* into lifecycle states: content is created, may go viral, eventually fatigues, and may be moderated. This content-centric framing works at the aggregate level where platform data is most available (retweet counts, reply volumes, and moderation actions are all content-level observables), separates amplification (which acts on content visibility) from user response (captured by a single aggregate user pool), and lets us study the platform health tradeoff directly through content lifecycle dynamics rather than individual user psychology.

The tradeoff is that we lose resolution on user heterogeneity and network topology. We treat users as a single aggregate pool U with constant replenishment and pressure-dependent attrition. This keeps the system tractable (six states rather than dozens) while still capturing the feedback loop between content amplification, discussion pressure, and platform health.

1.4 The dataset at a glance

We use the SNAP Higgs Twitter dataset [2] as our main data source: a record of approximately 563,000 social interactions captured during a 168-hour observation window centred on the July 2012 Higgs boson announcement. The dataset records three event types, namely retweets (RT), replies (RE), and mentions (MT), with Unix-epoch timestamps at one-second resolution. Figure 1 shows the hourly activity for each type. The dominant feature is a sharp retweet cascade in the first 24 hours, followed by a slower decay. Reply activity, which we later use as a proxy for discussion pressure, tracks the retweet peak with a slight lag. Mention activity is noisier and lower in volume.

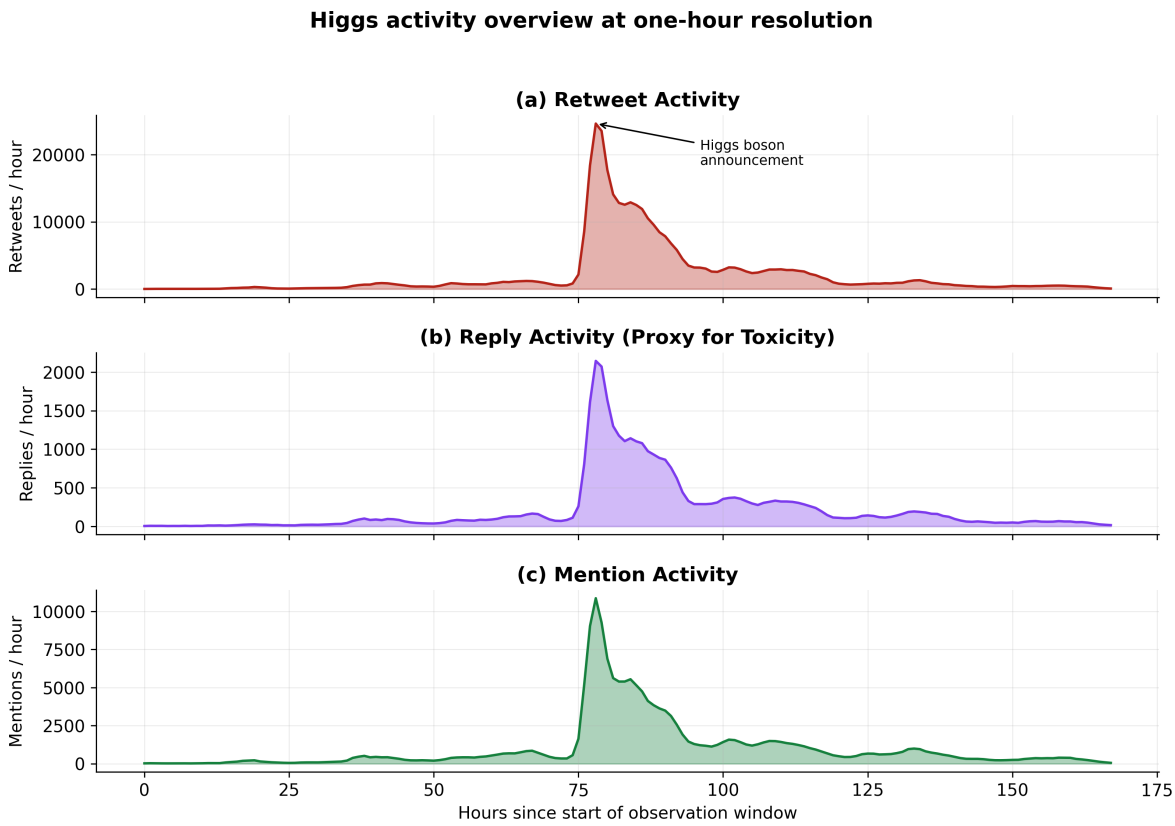

Figure 1: Hourly retweet, reply, and mention counts in the SNAP Higgs Twitter dataset over the full 168-hour observation window. The retweet panel isolates the dominant burst used for spread calibration, while the reply panel tracks the same event with a high-arousal discussion signal that we use as the observable proxy for latent toxicity. The figure matters because it shows, directly from the raw data, why the project separates spread calibration from reply-driven toxicity calibration.

1.5 Operational definition of rage-bait

We define *rage-bait content* operationally as any content that generates disproportionate reply and engagement activity relative to its passive resharing volume, captured in our model by the ratio of discussion pressure τ to viral volume V. This is a model-level definition, not a colloquial one: content is "rage-bait" whenever the pressure-to-virality ratio is elevated, regardless of the topic's political valence or emotional register. The Higgs boson announcement satisfies this definition empirically. During the main cascade, reply and mention activity surges well beyond what the retweet volume alone would predict (Figure 1b and c), consistent with high-arousal content dynamics in which the event provokes intense discussion, not just passive resharing. So the Higgs cascade works as a demonstration dataset for the IVFS model: it exhibits the structural signature, a high τ/V ratio, that the model is designed to capture, even though the underlying content is scientific rather than politically provocative.

Throughout the report, when we use the word *toxicity*, we mean this same latent rage bait toxicity: the reply driven discussion pressure represented by τ and by its feedback into spread, fatigue, and user attrition. This is an aggregate behavioural quantity inferred from Higgs reply activity, not a direct text label. When we specifically mean *Jigsaw toxicity*, we mean the external text level annotation scores from the Jigsaw dataset, which are used only as an outside reference

for scale.

Throughout this report we return to one central question: how does algorithmic amplification change the long-run balance between virality, discussion pressure, and user retention in a content-centric social-media system? The sections that follow build the model, establish the threshold, calibrate it against data, and trace the answer through to a quantitative policy comparison.

2 Previous mathematical models

We review the most relevant prior models and where our IVFS model differs.

2.1 Rumour models and epidemic analogies

The idea that information spreads like a disease dates to Daley and Kendall [3], who proposed a stochastic model for rumour transmission in which "spreaders" contact "ignorants" and either transmit the rumour or become "stiflers." This model has a natural compartmental structure analogous to SIR, and its threshold behaviour (rumours persist only when the contact rate exceeds a critical value) mirrors the basic reproduction number $\mathcal{R}_0$ in epidemiology.

De Domenico et al. [2] applied epidemic-style modelling to the same Higgs Twitter dataset that we use. They studied the temporal evolution of the retweet cascade around the Higgs boson discovery announcement and fitted several spreading models to the data. Their work demonstrates that compartmental models can capture the burst-and-decay dynamics of real social media cascades, which is one of the empirical motivations for our project.

2.2 User-centric compartmental models

Several recent papers have developed compartmental ODE models specifically for social media dynamics, but with *users* as the compartmentalized entities.

Li and Shao [8] proposed an SVFR (Susceptible-Viewing-Forwarding-Recovered) model for viral video sharing. In their framework, users transition from viewing to forwarding based on content attractiveness, and advertising interventions can modify transition rates. The model is fitted to video-sharing data and produces threshold conditions for viral spread. Jing et al. [9] developed a similar SIR-based model for advertising spread on social networks, incorporating network structure through degree distributions.

These models give useful threshold results, but they assume user-level states that are hard to observe at platform scale. For example, classifying a user as "susceptible" or "recovered" requires knowledge of their exposure history, which is typically not available in public datasets.

2.3 Toxicity propagation models

A separate line of work models the propagation of toxic or harmful content on social media using epidemic-style compartmental structures.

Maleki et al. [10] applied an epidemiological model to evaluate toxicity propagation on Twitter. Their model classifies users into compartments based on their toxicity exposure and production, and they calibrate it using text-based toxicity scores from Twitter data. Addai et al. [11] extended this approach by modelling interventions (content moderation, counter-speech, and user education) as control variables in a toxicity-propagation ODE system and studying their effects through simulation.

Both of these models use direct text-based toxicity labels, which is an advantage over our proxy-based approach. However, both also compartmentalize *users* rather than content, and neither explicitly models the feedback loop between content amplification, discussion pressure, and user retention that is central to our model.

2.4 Positioning of the present model

Table 1 summarizes the key distinctions. Our IVFS model differs from the prior work in three respects: (i) it compartmentalizes content lifecycle states rather than user behavioural states, (ii) it explicitly couples content dynamics to a latent discussion-pressure variable and an active-user pool, and (iii) it is calibrated on a real event cascade (the Higgs dataset) rather than on simulated or aggregate data alone.

Table 1: Comparison of compartmental modelling approaches for social media dynamics.

Model	Compartments	Pressure	User feedback	Calibration
Daley-Kendall [3]	Users	No	No	Analytical
Li-Shao SVFR [8]	Users	No	No	Video sharing data
Jing et al. SIR [9]	Users	No	No	Simulated
Maleki et al. [10]	Users	Toxicity labels	No	Twitter toxicity
Addai et al. [11]	Users	Toxicity labels	No	Simulated
This paper (IVFS)	**Content**	**Reply proxy**	**Yes**	**Higgs retweet data**

The main difference is the content centric perspective. Since we track content through lifecycle stages rather than users through behavioural stages, the state variables correspond directly to observable platform quantities (content volumes, reply rates, active user counts) rather than latent user states that would need to be inferred. Our toxicity variable τ is driven by reply volume, the same behavioural signal the Higgs dataset provides, rather than by text level toxicity labels.

3 Modelling question

Our main question is:

> *How does platform amplification shape the full lifecycle of rage bait content, from creation through virality, fatigue, and suppression, and what are the downstream consequences for toxicity and long run user retention?*

We investigate five subsidiary questions:

1. At what amplification level does viral spread become self-sustaining, and what is the mathematical nature of this threshold (transcritical bifurcation, forward or backward)?
2. How does the toxicity feedback loop (where viral content drives reply activity and latent toxicity, which in turn amplifies further spread) affect the equilibrium structure of the system?
3. How sensitive are the long-run equilibrium outcomes (V^*, τ^*, U^*) to the parameters governing the pressure block, and which parameters matter most?

4. Can a single dataset (the SNAP Higgs Twitter activity stream) support both a spread calibration and a toxicity proxy, using reply activity as the observable signal of rage bait toxicity?

5. What is the quantitative tradeoff between platform engagement and platform health, and is there an amplification level that balances the two?

These questions fit a compartmental ODE framework: they involve thresholds, equilibria, bifurcation, and sensitivity, all standard tools from the course. The lifecycle framing means we track content through its full trajectory on the platform, not just the initial viral spike.

4 Modelling approach

We use a deterministic six-state compartmental ordinary differential equation (ODE) system. In the course's classification, this is a *deterministic dynamical model*: the state evolves continuously in time according to a system of first-order ODEs, and there is no stochastic or probabilistic component.

The modelling approach follows the DABAR procedure:

1. **Define.** We define the modelling question around the amplification and health tradeoff, with content lifecycle states as the primary dynamical variables.

2. **Assume.** We assume that aggregate content and user activity can be approximated by continuous deterministic flows, that content items can be grouped into four lifecycle compartments, and that harmful discussion pressure can be summarized by a single latent scalar state.

3. **Build.** We construct a six-state ODE system with content compartments (I, V, F, S), a pressure state τ, and a user pool U, with couplings that capture the amplification feedback loop.

4. **Assess.** We calibrate the spread block against the Higgs Twitter dataset, benchmark against a classical SIR alternative, validate on multiple time windows, and perform sensitivity and robustness analyses.

5. **Report.** We present the results with clear attention to what is well-identified (spread parameters) and what is only partially identified (pressure parameters).

The empirical strategy has two stages. First, we calibrate the spread block using a reduced IVF helper model fitted to the Higgs retweet stream. This stage identifies the spread-side parameters $(\beta_0, \gamma_0, \lambda_0, d_\lambda)$ with narrow confidence intervals. Second, we simulate the full six-state IVFS model under different amplification scenarios to study the policy tradeoff. The pressure block is handled separately: same-dataset reply activity (RE) serves as a proxy for latent discussion pressure, and the Jigsaw toxicity dataset provides an external reference scale for the magnitude of τ, but neither source uniquely pins down the pressure parameters (ϕ, ψ).

This split means the spread calibration is clean and well-identified, while the pressure block is reported as partially constrained. This is better than forcing a joint calibration that would mix well-identified and poorly-identified parameters.

5 Formulation of the model

5.1 State variables

The IVFS model is a deterministic compartmental ODE system with six state variables. Four are content lifecycle compartments, one captures latent discussion pressure, and one tracks the

aggregate active-user pool.

Variable	Interpretation
$I(t)$	Ignored or not-yet-viral content. This compartment represents content items that have been posted on the platform but have not yet achieved significant visibility. Content enters the system through I.
$V(t)$	Viral content. Content items that have achieved high visibility and are being actively shared and engaged with. This is the compartment most directly affected by the amplification parameter α.
$F(t)$	Fatigued content. Content that was previously viral but has lost engagement momentum. Users have moved on, and the content is no longer being actively shared.
$S(t)$	Suppressed or moderated content. Content that has been removed or de-ranked by platform moderation. In the current model, moderation acts only on ignored content ($I \to S$).
$\tau(t)$	Latent toxicity of rage baiting. A scalar state variable representing the aggregate level of reply driven, high arousal discussion pressure on the platform. In the calibration, τ is proxied by same dataset reply volume (RE events from Higgs), which is the behavioural signal most directly consistent with rage bait dynamics: content that provokes replies rather than passive reshares. This is not directly observed text toxicity; it is a latent variable driven by viral content and relaxing over time.
$U(t)$	Active users. The aggregate number of users actively participating on the platform. This pool is replenished at a constant rate and depleted at a rate that increases with discussion pressure.

The four content compartments (I, V, F, S) form a lifecycle: content is created (enters I), may go viral (moves to V), eventually fatigues (moves to F), and may be moderated at any point (moves to S). All content also decays naturally at rate μ_c, representing loss of visibility over time. This lifecycle structure is the core of the model and is what distinguishes it from user-centric approaches.

5.2 Parameters

Table 2 lists all parameters in the model. We distinguish between *calibrated parameters*, which are fitted to data, and *scenario parameters*, which are fixed at values chosen to represent different policy configurations.

Table 2: IVFS model parameters.

Parameter	Value	Type	Interpretation
α	varies	Policy	Amplification level (platform tuning knob)
β_0	1.000073	Calibrated	Base spread rate (fitted from Higgs RT)
γ_0	0.300283	Calibrated	Base fatigue rate (fitted from Higgs RT)
λ_0	0.199996	Calibrated	Exogenous seed amplitude
d_λ	0.044641	Calibrated	Exogenous seed decay rate
κ	0.8	Scenario	Pressure effect on amplification
η	0.3	Scenario	Pressure effect on fatigue
ϕ	0.056	Scenario	Viral-to-pressure forcing rate
ψ	0.1	Scenario	Pressure relaxation rate
ρ	0.06	Scenario	Content injection rate per user
λ_u	0.02	Scenario	User attrition rate
ν	1.0	Scenario	User replenishment rate
μ_c	0.01	Scenario	Natural content decay rate
δ	0.05	Scenario	Moderation / suppression rate
w	10.0	Scenario	Pressure effect on user attrition

The amplification parameter $\alpha \in [0, 1]$ is the key policy variable. It represents the degree to which the platform's recommendation algorithm promotes engaging content. Setting α close to 1 corresponds to aggressive engagement optimization; setting α close to 0 corresponds to a health-first policy that limits algorithmic amplification.

5.3 Modelling assumptions

The assumptions in the IVFS model are not arbitrary simplifications added for convenience. Each one is a modelling choice tied to the level of data available and to the specific policy question we want to answer. The Higgs dataset gives aggregate event streams, not individual exposure histories, so the right standard is whether an assumption is appropriate for aggregate platform scale dynamics. With that in mind, the model uses the following assumptions:

1. **Continuous deterministic flows.** Aggregate content and user activity can be approximated by continuous variables evolving according to deterministic ODEs. This is appropriate here because the calibration window contains very large event counts, so the main object of interest is the mean cascade shape rather than individual retweet randomness. The deterministic approximation is therefore a deliberate coarse graining step, not an attempt to model micro-level behaviour.

2. **Content lifecycle compartments.** Content items can be meaningfully grouped into four lifecycle stages: ignored, viral, fatigued, and suppressed. These boundaries are stylized, but they are exactly the level at which platform policy acts. Platforms tune visibility, moderation, and ranking at the content level, so a content-centric partition is the right abstraction for studying amplification policy.

3. **Homogeneous mixing.** We assume that all content items in a given compartment have equal probability of transitioning to other compartments. This loses network topology and community structure, but it is what allows the model to answer a platform-level control question with public

aggregate data. A network-aware model would require information that the present dataset does not contain.

4. **Scalar pressure.** Harmful discussion dynamics are summarized by a single scalar state τ. This does not claim that all forms of online toxicity are the same. It claims only that, for the present question, a one-dimensional latent toxicity variable is enough to capture the feedback loop between virality, heated reply activity, and user loss.

5. **Aggregate user pool.** The active-user population is a single aggregate pool with constant replenishment and pressure-dependent attrition. This is justified because the report asks how platform health responds in the aggregate. The variable U is not intended to resolve user types; it is intended to detect the directional retention cost of amplification.

6. **Local parameter constancy.** The spread-side parameters (β_0, γ_0) are treated as locally constant on the 100-hour calibration window. This is an identification choice: we calibrate the dominant burst and immediate decay on a fixed event window, then check robustness by refitting alternate windows and coarser binning choices rather than pretending the same coefficients must hold identically over the full week.

The common theme is that each assumption trades granularity for identifiability. What we learn from this section is that the model is deliberately matched to the resolution of the available data and to the platform-level policy question, even though that leaves important micro-level effects for future work.

5.4 The IVFS equations

The model is the following system of six coupled first-order ODEs:

$$\dot{I} = \underbrace{\frac{\rho U}{1+U}}_{\text{content creation}} - \underbrace{\beta_{\text{eff}}\, I\, V}_{\text{viral conversion}} - \underbrace{(\delta + \mu_c)\, I}_{\text{moderation + decay}}, \tag{1}$$

$$\dot{V} = \underbrace{\beta_{\text{eff}}\, I\, V}_{\text{viral conversion}} - \underbrace{\gamma_{\text{eff}}\, V}_{\text{fatigue}} - \underbrace{\mu_c\, V}_{\text{decay}}, \tag{2}$$

$$\dot{F} = \underbrace{\gamma_{\text{eff}}\, V}_{\text{fatigue inflow}} - \underbrace{\mu_c\, F}_{\text{decay}}, \tag{3}$$

$$\dot{S} = \underbrace{\delta\, I}_{\text{moderation}} - \underbrace{\mu_c\, S}_{\text{decay}}, \tag{4}$$

$$\dot{\tau} = \underbrace{\phi\, V}_{\text{pressure forcing}} - \underbrace{\psi\, \tau}_{\text{pressure relaxation}}, \tag{5}$$

$$\dot{U} = \underbrace{\nu}_{\text{replenishment}} - \underbrace{\lambda_u (1 + w\tau)\, U}_{\text{pressure-dependent attrition}}, \tag{6}$$

where the effective rates are

$$\beta_{\text{eff}} = \alpha\, \beta_0\, (1 + \kappa\, \tau), \qquad \gamma_{\text{eff}} = \gamma_0\, (1 + \eta\, \tau). \tag{7}$$

Each equation works as follows.

Content creation (Eq. 1). New content enters the ignored compartment at rate $\rho U/(1+U)$. This is a saturating (Michaelis-Menten type) inflow: when the user pool U is small, content creation scales roughly linearly with U; when U is large, creation saturates toward the maximum rate ρ. A purely linear inflow ρU made the content input unrealistically large in our simulations, which is why we adopted the saturating form. Content leaves the ignored compartment either by going viral (at rate $\beta_{\text{eff}} IV$, which is proportional to the product of ignored and viral content, analogous to the mass-action transmission term in SIR models) or by being moderated (δI) or decaying naturally ($\mu_c I$).

Viral dynamics (Eq. 2). Content enters the viral compartment through the mass-action conversion term $\beta_{\text{eff}} IV$. The effective spread rate $\beta_{\text{eff}} = \alpha\beta_0(1+\kappa\tau)$ captures two effects: the platform amplification parameter α scales the base rate β_0, and the pressure feedback $(1+\kappa\tau)$ means that higher discussion pressure increases the rate at which content goes viral. This is the amplification feedback loop. Viral content leaves V by fatiguing at rate $\gamma_{\text{eff}} V$ or decaying at rate $\mu_c V$.

Fatigue and suppression (Eqs. 3 and 4). These are accumulation compartments. Fatigued content collects previously viral items that have lost momentum; suppressed content collects items removed by moderation. Neither F nor S feeds back into the spread-pressure-user dynamics. They are "sinks" in the feedback structure, but they are important for interpreting the content lifecycle and for total content accounting.

Discussion pressure (Eq. 5). The latent pressure state τ is driven by viral content at rate ϕV and relaxes at rate $\psi\tau$. This is a simple first-order linear response: pressure builds when viral content is present and decays exponentially when it is not. The equilibrium pressure level (when V is constant) is $\tau^* = \phi V^*/\psi$, so the ratio ϕ/ψ sets the scale of long-run pressure.

User dynamics (Eq. 6). Users join the platform at a constant rate ν and leave at rate $\lambda_u(1+w\tau)U$. The important part is pressure-dependent attrition: higher discussion pressure causes faster user departure. At equilibrium, $U^* = \nu/[\lambda_u(1+w\tau^*)]$, so increased pressure directly reduces the steady-state user base.

Effective rates (Eq. 7). The effective spread rate β_{eff} and effective fatigue rate γ_{eff} both depend on the pressure τ. Higher pressure increases both rates: it makes content spread faster (viral amplification) but also fatigue faster (burnout effect). This creates a nonlinear feedback: amplification drives viral content, which drives pressure, which in turn modifies both the amplification and fatigue rates.

5.5 Structure of the feedback loop

Figure 2 shows the state-flow structure of the model. The essential feedback loop is:

$$V \xrightarrow{\phi} \tau \xrightarrow{\kappa} \beta_{\text{eff}} \to V.$$

Viral content drives pressure, pressure amplifies the effective spread rate, and the amplified spread rate produces more viral content. This positive feedback loop is counteracted by fatigue ($V \to F$), pressure relaxation ($\psi\tau$), and pressure-dependent user attrition ($\lambda_u w\tau U$), which reduces the content creation rate.

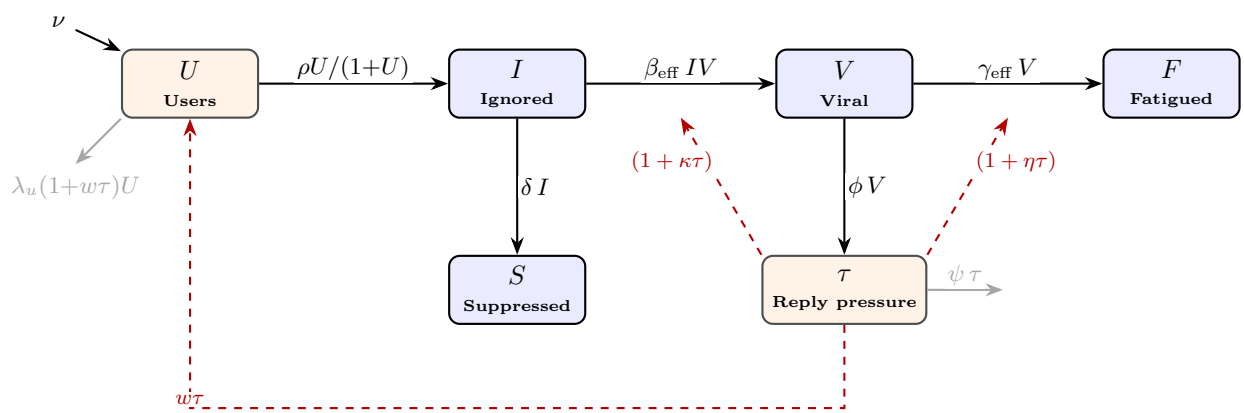

Figure 2: Mechanism diagram for the six-state IVFS model. The figure shows the content lifecycle $I \to V \to F$ with moderation diverting flow to S, together with the latent toxicity state τ and the active-user pool U. The central feedback loop is visible in one picture: viral content raises toxicity, toxicity modifies spread and fatigue, and toxicity also lowers user retention. This is the core mechanism of the paper, so the diagram matters because it makes the amplification-toxicity-to-retention tradeoff interpretable before any equations or calibration results are read.

5.6 Threshold quantity and disease-free equilibrium

A central quantity in any compartmental model is the threshold that determines whether the "infection" (in our case, viral content) can persist in the long run. Following standard practice in epidemic modelling, we compute this threshold by linearizing around the disease-free equilibrium (DFE).

At the DFE, there is no viral content: $V = 0$, $\tau = 0$, $F = 0$. The remaining equations give:

$$U_{\text{DFE}} = \frac{\nu}{\lambda_u}, \qquad I_{\text{DFE}} = \frac{\rho\, U_{\text{DFE}}}{(1 + U_{\text{DFE}})(\delta + \mu_c)}. \tag{8}$$

At the DFE, the suppressed-content compartment satisfies $\dot{S} = \delta I - \mu_c S = 0$, giving

$$S_{\text{DFE}} = \frac{\delta\, I_{\text{DFE}}}{\mu_c}.$$

With the current parameter values, $U_{\text{DFE}} = 50.0$, $I_{\text{DFE}} \approx 0.9804$, and $S_{\text{DFE}} = 0.05 \times 0.9804/0.01 \approx 4.902$.

Linearizing the $\dot{V}$ equation around the DFE yields

$$\dot{V} \approx \big(\alpha\, \beta_0\, I_{\text{DFE}} - \gamma_0 - \mu_c\big) V.$$

Viral content grows when the coefficient is positive and decays when it is negative. We define the basic reproduction number:

$$\boxed{\mathcal{R}_0 = \frac{\alpha\, \beta_0\, I_{\text{DFE}}}{\gamma_0 + \mu_c}.} \tag{9}$$

Viral content is self-sustaining when $\mathcal{R}_0 > 1$ and dies out when $\mathcal{R}_0 < 1$. Solving $\mathcal{R}_0 = 1$ for α gives the critical amplification level:

$$\alpha_{\text{crit}} = \frac{\gamma_0 + \mu_c}{\beta_0\, I_{\text{DFE}}} \approx 0.3165. \tag{10}$$

This threshold divides the amplification parameter space into two regimes:

- $\alpha < \alpha_{\text{crit}}$: the DFE is stable, viral content dies out, and $\tau \to 0$.
- $\alpha > \alpha_{\text{crit}}$: the DFE is unstable, and a positive (endemic) equilibrium with $V^* > 0$ and $\tau^* > 0$ exists.

This $\mathcal{R}_0$ comes from linearizing the $\dot{V}$ equation at the DFE. A full next-generation matrix derivation would strengthen this result; we discuss this as a future extension in Section 11.3.

The mechanism the model proposes is a nonlinear amplification-pressure feedback loop: platform amplification α converts ignored content to viral, viral content builds latent discussion pressure, and that pressure boosts further spread while eroding the user base. Whether this feedback self-sustains depends entirely on whether α exceeds α_{crit}.

In the numerical implementation, trajectories are integrated with SciPy's adaptive 'solve_ivp' solver using explicit tolerances and a one-hour maximum step. The code also clamps small negative state excursions back to zero inside the right-hand side evaluation. This does not change the analytical model; it is a numerical safeguard that keeps the computed trajectory inside the nonnegative region established above.

6 Analysis of the model

This section presents the analytical results we have obtained for the IVFS system. We establish nonnegativity, boundedness, the DFE threshold, local stability, and sensitivity indices.

6.1 Nonnegativity

Proposition 1 (Nonnegativity)**.** *If the initial state satisfies $I(0), V(0), F(0), S(0), \tau(0), U(0) \geq 0$, then the solution remains componentwise nonnegative for as long as it exists.*

Proof. We verify that each coordinate hyperplane is inward-pointing or tangent.

- At $I = 0$: $\dot{I} = \rho U/(1+U) \geq 0$, since $U \geq 0$.
- At $V = 0$: $\dot{V} = 0$, since both the conversion and fatigue terms vanish.
- At $F = 0$: $\dot{F} = \gamma_{\text{eff}} V \geq 0$, since $V \geq 0$ and $\gamma_{\text{eff}} > 0$.
- At $S = 0$: $\dot{S} = \delta I \geq 0$, since $I \geq 0$ and $\delta > 0$.
- At $\tau = 0$: $\dot{\tau} = \phi V \geq 0$, since $V \geq 0$ and $\phi > 0$.
- At $U = 0$: $\dot{U} = \nu > 0$, since $\nu > 0$.

Since the vector field points inward (or is tangent) on every face of the nonnegative orthant $\mathbb{R}^6_{\geq 0}$, the orthant is positively invariant. □

6.2 Boundedness

Proposition 2 (Bounded invariant region)**.** *Solutions of the IVFS system with nonnegative initial data enter a bounded positively invariant region in finite time.*

Proof. We bound each group of variables in turn.

User pool. From (6), $\dot{U} = \nu - \lambda_u(1 + w\tau)U \leq \nu - \lambda_u U$, since $\tau \geq 0$. By the comparison lemma, $U(t) \leq \max\{U(0), \nu/\lambda_u\}$ for all $t \geq 0$. Denote $U_{\max} = \nu/\lambda_u$.

Total content mass. Define $C = I + V + F + S$. Summing equations (1) to (4), the internal transfer terms ($\beta_{\text{eff}} IV$, $\gamma_{\text{eff}} V$, δI) cancel, leaving:

$$\dot{C} = \frac{\rho U}{1 + U} - \mu_c C \leq \rho - \mu_c C.$$

Hence $C(t) \leq \max\{C(0), \rho/\mu_c\}$. Denote $C_{\max} = \rho/\mu_c$.

Pressure. Since $V \leq C \leq C_{\max}$, equation (5) gives $\dot{\tau} \leq \phi C_{\max} - \psi\tau$, so $\tau(t) \leq \max\{\tau(0), \phi C_{\max}/\psi\}$.

All six state variables are therefore bounded, and the bounds define a compact positively invariant region. □

Remark 1. *Propositions 1 and 2 together guarantee that the IVFS system is well-posed: solutions exist for all forward time, remain nonnegative, and stay bounded. This lets us do the equilibrium and stability analysis that follows.*

6.3 Disease-free equilibrium and threshold

Proposition 3 (DFE stability threshold)**.** *The disease-free equilibrium* $(I_{\text{DFE}}, 0, 0, S_{\text{DFE}}, 0, U_{\text{DFE}})$ *is locally asymptotically stable when* $\mathcal{R}_0 < 1$ *and unstable when* $\mathcal{R}_0 > 1$*, where* $\mathcal{R}_0$ *is given by* (9)*.*

Proof. At the DFE, $V = \tau = 0$, and the Jacobian of the full system has a block-triangular structure. The eigenvalue governing the V direction is

$$\mu_V = \alpha\, \beta_0\, I_{\text{DFE}} - \gamma_0 - \mu_c = (\gamma_0 + \mu_c)(\mathcal{R}_0 - 1).$$

When $\mathcal{R}_0 < 1$, this eigenvalue is negative. The remaining five eigenvalues of the block-triangular Jacobian at the DFE are:

$$\mu_\tau = -\psi, \quad \mu_F = -\mu_c, \quad \mu_I = -(\delta + \mu_c), \quad \mu_S = -\mu_c, \quad \mu_U = -\lambda_u,$$

all of which are strictly negative since every parameter is positive. Hence the DFE is locally asymptotically stable. When $\mathcal{R}_0 > 1$, the V eigenvalue is positive and the DFE is unstable. □

The transition at $\mathcal{R}_0 = 1$ is consistent with a *transcritical bifurcation*: the DFE exchanges stability with a positive equilibrium as α passes through α_{crit}. Below the threshold, only the DFE exists (among nonnegative equilibria). Above the threshold, a unique positive equilibrium with $V^* > 0$ bifurcates from the DFE and is locally stable. This is the standard bifurcation structure for epidemic-type compartmental models; a formal center-manifold coefficient calculation confirming the forward direction remains as future work.

Figure 3 shows the bifurcation numerically. Panel (a) plots $\mathcal{R}_0$ as a function of α, with the threshold at $\mathcal{R}_0 = 1$ clearly marked. Panel (b) shows the equilibrium viral volume V^* computed by integrating the full ODE to steady state for each value of α. The sharp onset at $\alpha_{\text{crit}} \approx 0.3165$ and the subsequent growth of V^* illustrate the transcritical bifurcation.

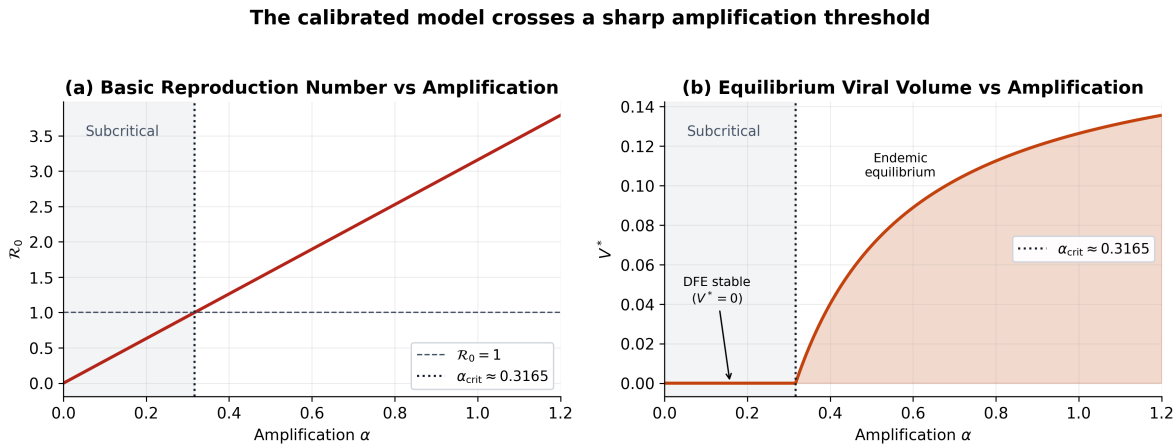

Figure 3: Threshold evidence from the calibrated IVFS model. Panel (a) plots the basic reproduction number $\mathcal{R}_0$ against amplification strength α, and panel (b) plots the numerically computed equilibrium viral volume V^* over the same sweep. The sharp onset near $\alpha_{\text{crit}} \approx 0.3165$ makes the transcritical threshold visually explicit, which matters because the platform-health tradeoff in later sections only appears once amplification crosses this critical level.

6.4 Positive equilibrium

When $\mathcal{R}_0 > 1$, the positive equilibrium can be computed by setting all time derivatives to zero. The pressure equation (5) gives $\tau^* = \phi V^*/\psi$, and the user equation (6) gives $U^* = \nu/[\lambda_u(1+w\tau^*)]$. Substituting into the content equations yields a system that can be solved numerically for (I^*, V^*).

6.5 Local stability at the positive equilibrium

The repository numerically evaluates the Jacobian of the full six-state system at the positive equilibrium for each scenario. The current numerical results are:

- At $\alpha = 0.2$ ($\mathcal{R}_0 \approx 0.632$): subcritical, no positive equilibrium exists.
- At $\alpha = 0.5$ ($\mathcal{R}_0 \approx 1.580$): all eigenvalues of the Jacobian at the positive equilibrium have negative real part. The equilibrium is locally asymptotically stable.
- At $\alpha = 0.9$ ($\mathcal{R}_0 \approx 2.844$): all eigenvalues of the Jacobian at the positive equilibrium have negative real part. The equilibrium is locally asymptotically stable.

The leading (least negative) real part of the eigenvalues at the positive equilibrium is approximately -0.01, which corresponds to the slow natural content decay eigenvalue from the decoupled sink compartments F and S.

A fully symbolic proof of local asymptotic stability of the positive equilibrium is beyond the scope of this report, because the six-dimensional Jacobian does not factor into small enough blocks for clean hand computation. However, the numerical Jacobian checks are consistent across all tested parameter values and support the claim that the positive equilibrium is stable whenever it exists.

6.6 Sensitivity analysis

We now check how sensitive the model outputs are to parameter changes. Following the course material, we define the *normalized sensitivity index* of a quantity x with respect to a parameter p as:

$$S(x,p) = \frac{\partial x}{\partial p} \cdot \frac{p}{x}. \tag{11}$$

This measures the proportional change in x per proportional change in p.

We compute the sensitivity of $\mathcal{R}_0$ with respect to the key parameters analytically. Since

$$\mathcal{R}_0 = \frac{\alpha\,\beta_0\,I_{\mathrm{DFE}}}{\gamma_0 + \mu_c},$$

and I_{DFE} depends on ρ, δ, μ_c, ν, and λ_u through (8), we obtain:

$$\begin{aligned} S(\mathcal{R}_0, \alpha) &= +1, \\ S(\mathcal{R}_0, \beta_0) &= +1, \\ S(\mathcal{R}_0, \gamma_0) &= -\frac{\gamma_0}{\gamma_0 + \mu_c} \approx -0.968. \end{aligned}$$

In practice: $\mathcal{R}_0$ is directly proportional to both the amplification level α and the base spread rate β_0 (unit elasticity), and nearly inversely proportional to the base fatigue rate γ_0. The critical threshold α_{crit} therefore shifts proportionally with γ_0/β_0: faster fatigue raises the bar for self-sustaining virality, and faster base spread lowers it.

For the long-run equilibrium quantities V^*, τ^*, and U^*, we compute sensitivity indices numerically by perturbing each parameter by $\pm 1\%$ and measuring the proportional response. The numerical sensitivity table is reported in Table 3.

Table 3: Numerical sensitivity indices for equilibrium quantities at the moderate scenario ($\alpha = 0.5$). Each entry is $S(x,p)$ computed by central finite differences with $\pm 1\%$ perturbation.

Parameter	$S(V^*,p)$	$S(\tau^*,p)$	$S(U^*,p)$
α	$+1.67$	$+1.67$	-0.469
β_0	$+1.67$	$+1.67$	-0.469
γ_0	-2.59	-2.59	$+0.726$
η	-0.03	-0.03	$+0.008$
κ	$+0.05$	$+0.05$	-0.014
ϕ	≈ 0	$+1.00$	-0.281
ψ	≈ 0	-1.00	$+0.281$
w	-0.02	-0.02	-0.275

Some indices exceed one in magnitude. Above the transcritical bifurcation, the nonlinear feedback amplifies small parameter perturbations, so a 1% change in a parameter can shift the equilibrium by more than 1%.

The main takeaways are:

1. The equilibrium viral volume V^* is most sensitive to γ_0 ($S \approx -2.59$), followed by α and β_0 ($S \approx +1.67$ each). These are all spread-side parameters, which confirms that the spread calibration is the most important empirical anchor.

2. The pressure parameters ϕ and ψ have essentially no effect on V^* but have unit-elastic effects on τ^* ($S \approx \pm 1.00$). This makes sense: at equilibrium, $\tau^* = \phi V^*/\psi$, so τ^* is directly proportional to ϕ and inversely proportional to ψ, while V^* is determined primarily by the spread-side balance.

3. The user pool U^* shows moderate sensitivity to the pressure parameters ($S \approx -0.28$ for ϕ, $+0.28$ for ψ) and to the pressure-attrition coupling w ($S \approx -0.28$). This reflects the pressure-dependent user attrition channel: more pressure drives more users away.

4. The pressure-amplification coupling κ has surprisingly small sensitivity ($S \approx 0.05$), indicating that the direct pressure-to-spread feedback is a second-order effect relative to the base spread parameters.

Figure 4 visualizes these sensitivity indices as bar charts, making the dominance of the spread-side parameters γ_0, α, and β_0 immediately apparent.

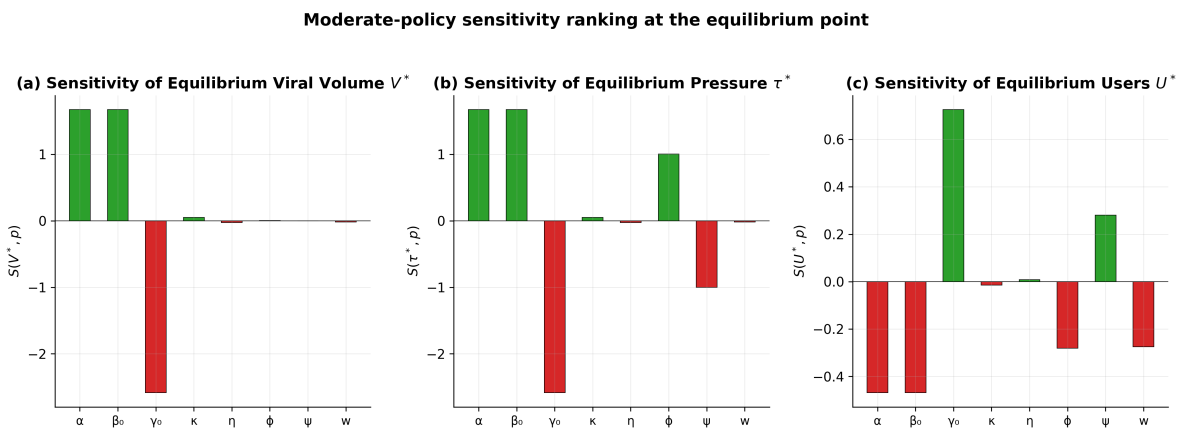

Figure 4: Normalized sensitivity indices for the equilibrium outputs V^*, τ^*, and U^* under the moderate policy scenario $\alpha = 0.5$. Positive bars indicate that increasing a parameter raises the corresponding equilibrium quantity, while negative bars indicate the opposite. The figure shows that spread-side parameters dominate virality and toxicity, whereas pressure-side parameters matter most for user retention, which is important for interpreting which levers are empirically identified and which levers drive the health tradeoff.

The structural insight from the analysis is that spread-side and pressure-side parameters govern nearly orthogonal outcomes: γ_0, α, and β_0 control viral volume and discussion pressure, while ϕ, ψ, and w primarily determine user retention. These are two largely independent levers available to a platform choosing where to intervene.

7 Data and calibration

7.1 The Higgs Twitter dataset

Our primary dataset is the SNAP Higgs Twitter activity stream [1, 2]. This dataset records social interactions on Twitter during the period surrounding the announcement of the Higgs boson discovery in July 2012. Each row in the activity file is a tuple of the form

`userA  userB  timestamp  interaction`,

where the interaction type is one of RT (retweet), RE (reply), or MT (mention). The dataset contains no tweet text, only interaction metadata.

The activity counts in the dataset are:

- Total events: 563,069
- Retweets (RT): 354,930
- Replies (RE): 36,902
- Mentions (MT): 171,237

The retweet activity shows a single dominant spike at approximately hour 78, with a peak of 30,602 retweets in that hour. This corresponds to the period immediately following the major announcements at CERN. We use a 100-hour calibration window from hour 68 to hour 167, centered around this main spike, for fitting the spread model. The window is long enough to capture both the spike and the subsequent decay tail.

We use the Higgs dataset in two distinct ways:

1. **Spread calibration.** Retweet counts are used to calibrate the reduced IVF helper model (Section 7.2).
2. **Pressure proxy.** Reply counts are used as a same-dataset proxy for the latent discussion pressure τ (Section 7.6).

7.2 Reduced IVF helper model

We calibrate the spread block using a reduced three-compartment IVF helper model, which is the $\dot{I}$, $\dot{V}$, $\dot{F}$ subsystem of the full IVFS model with $\tau = 0$ (no pressure feedback) and a decaying exogenous seed term:

$$\lambda(t) = \lambda_0 \, e^{-d_\lambda t}.$$

This seed term accounts for the external injection of content into the Higgs cascade (e.g., from news coverage) and improves the fit of the decay tail. The helper also sets $\delta = 0$ (no moderation attrition); the full model's DFE and equilibrium calculations restore the correct $(\delta + \mu_c)$ attrition. The helper model has five free parameters: β_0, γ_0, λ_0, d_λ, and the initial viral volume V_0. The initial condition is normalised so that $I(0) + V(0) = 1$ with $F(0) = 0$; because the output $V(t)$ is subsequently normalised by its peak value, this choice does not affect the fitted parameters.

The optimization uses a bounded Powell search with warm starts, minimizing a weighted sum of squared errors that emphasizes the spike region. The fitted parameter values are:

$$\beta_0 = 1.000073, \qquad \gamma_0 = 0.300283,$$
$$\lambda_0 = 0.199996, \qquad d_\lambda = 0.044641.$$

7.3 Calibration diagnostics

The spread calibration diagnostics from the current implementation are:

- Weighted loss: 0.002286
- Full-window R^2: 0.9247
- Spike-window R^2: 0.8891

- Tail mean bias: +0.0043

The full-window $R^2 = 0.92$ indicates that the reduced IVF helper model explains over 92% of the variance in the normalized retweet series. The spike-window $R^2 = 0.89$ shows that the model captures the main burst well but not perfectly, which is expected given the deterministic approximation to an inherently stochastic cascade.

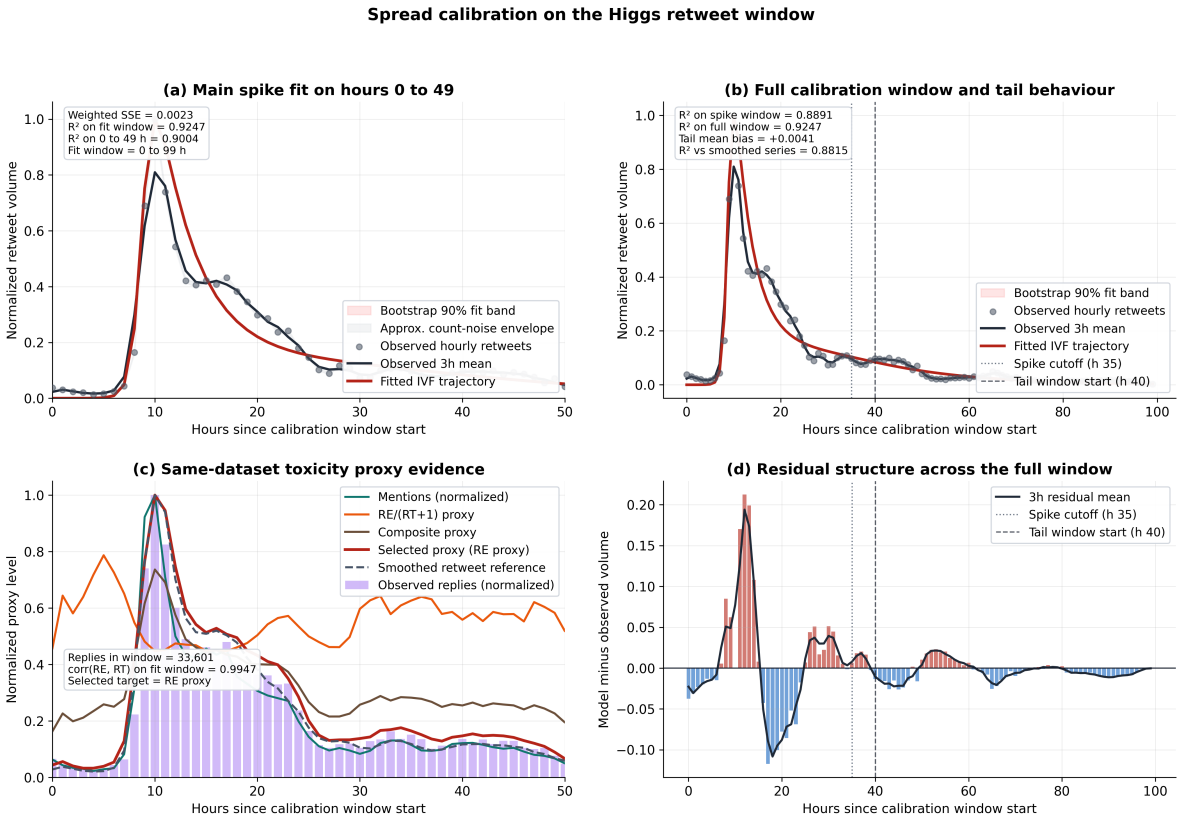

Figure 5: Calibration evidence for the main 100-hour Higgs retweet window. Panels (a) and (b) show observed hourly retweets, the fitted IVF curve, and a bootstrap fit band over the spike and over the full window; panel (c) compares the selected reply-based toxicity proxy with alternative same-dataset proxy constructions; panel (d) shows residual structure across time. The figure is intentionally honest about the tail mismatch: the spike is fitted tightly, while the later tail is harder because the deterministic helper compresses a noisier and lower-volume phase of the cascade. This matters because it separates what is strongly identified, namely the spread-side shape, from what is only approximate in the late decay.

7.4 Uncertainty quantification

We assess parameter uncertainty using two complementary methods.

Parametric bootstrap. We generate $B = 500$ bootstrap replicates by resampling the Higgs retweet counts from a Poisson distribution with mean equal to the model-fitted count at each time point (parametric bootstrap). Each replicate is refitted, producing a distribution of parameter

estimates. The resulting 95% confidence intervals are:

$$\beta_0 \in [0.9982,\ 1.0019],$$
$$\gamma_0 \in [0.2964,\ 0.3040],$$
$$\alpha_{\text{crit}} \in [0.3129,\ 0.3199].$$

The narrow intervals confirm that the spread parameters are practically identifiable from the Higgs data.

Profile likelihood. We compute local profile-likelihood curves by fixing one parameter at a grid of values near its optimum and re-optimizing the remaining parameters. Both β_0 and γ_0 show clear, well-defined minima (Figure 6), confirming practical identifiability.

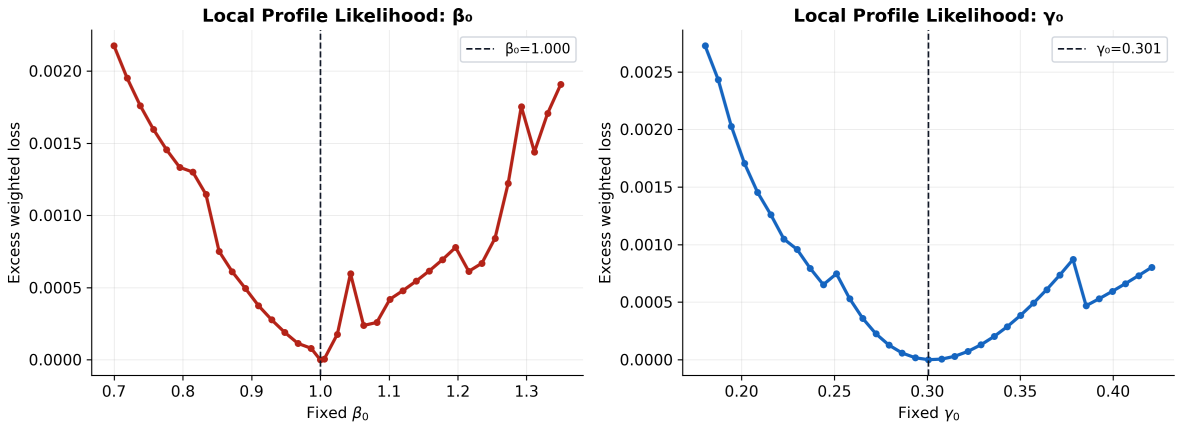

Figure 6: Profile likelihood curves for the fitted spread parameters β_0 and γ_0 on the Higgs calibration window. Each panel fixes one parameter over a grid and re-optimizes the remaining parameters, so the minimum indicates how sharply the data select the fitted value. The clear local minima support practical identifiability of the spread block, which matters because the threshold calculation depends directly on these two parameters.

7.5 Benchmark against a classical SIR model

To verify that the lifecycle structure in the IVFS model is empirically justified, we benchmark the reduced IVF helper model against a classical three-parameter SIR model fitted to the same normalized retweet window. On the spike-focused comparison window (Figure 7), the IVF fit achieves $R^2 \approx 0.889$ while the SIR alternative achieves $R^2 \approx 0.690$. The IVF model captures the asymmetric spike shape and the gradual decay tail significantly better than SIR, which supports the added lifecycle structure, especially the distinction between viral and fatigued content, as an empirically grounded modelling choice.

Just as important is what the benchmark cannot test. A classical SIR model has no state corresponding to latent toxicity and no user-retention block, so it cannot address the main policy question of this report. It can only ask whether a three-state epidemic curve can mimic the Higgs spike. The benchmark therefore validates the spread-side shape advantage of IVF over SIR, while the full IVFS model is needed for the toxicity and platform-health tradeoff itself.

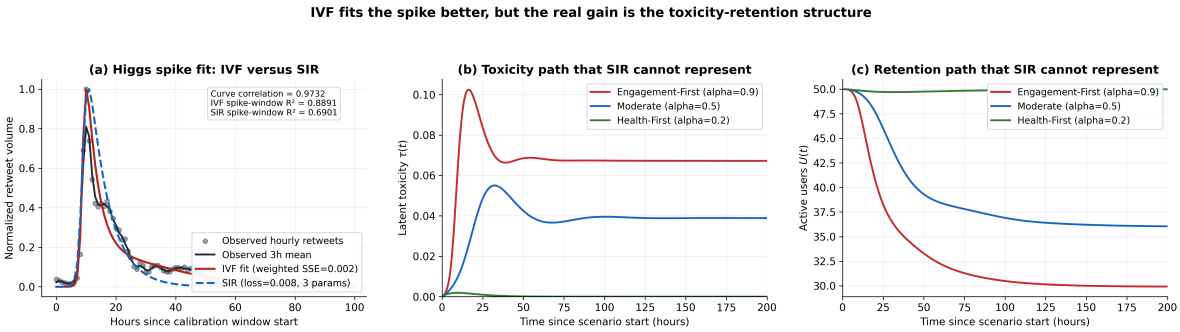

Figure 7: Benchmark comparison between the reduced IVF helper model and a classical three-parameter SIR alternative on the same Higgs retweet window. The left panel compares the empirical spike to both fitted curves, while the two right panels show toxicity and user-retention trajectories that only the IVFS framework can represent. The key takeaway is twofold: IVF fits the observed spike shape better than SIR, and the real scientific gain of IVFS is that it carries the calibration into a toxicity-retention tradeoff that SIR cannot test.

7.6 Observed proxy for toxicity

The interpretation of τ is one of the most important modelling decisions in this project. In this report, $\tau(t)$ is the latent toxicity of rage baiting, namely the reply driven discussion pressure generated by viral content and fed back into spread and user loss. The Higgs dataset contains no tweet text, so same dataset text toxicity cannot be measured directly. Instead, we use same dataset reply activity (RE) as the primary observable proxy for this toxicity.

Replies on social media are more likely to represent active discussion (including heated or contentious exchanges) than retweets (which are primarily a spreading mechanism) or mentions (which may be purely informational). High reply volume during a viral event is therefore a plausible indicator of elevated rage bait toxicity, even without text level toxicity labels.

We construct a proxy family and select the best-supported target:

- Smoothed RE proxy: $R^2 \approx 0.7422$ (unconstrained fit).
- Reply-rate proxy RE/(RT + 1): $R^2 \approx -1.14$ (poor fit).
- Composite proxy: $R^2 \approx -0.53$ (poor fit).

The RE proxy is the clear winner in terms of fit quality, and it is also the most directly interpretable.

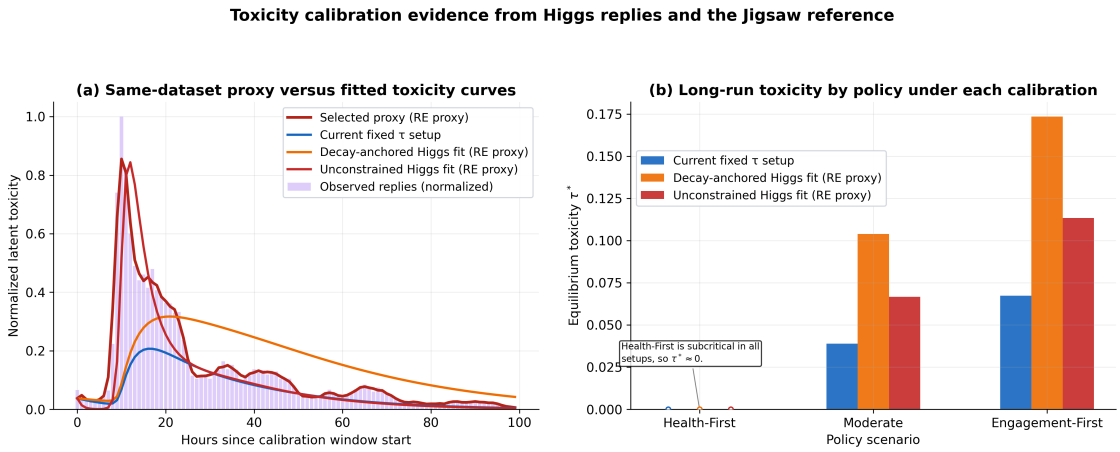

Figure 8: Toxicity-calibration evidence from Higgs replies and the external Jigsaw reference. Panel (a) compares the selected reply-based proxy with the baseline, decay-anchored, unconstrained, and reference-constrained toxicity trajectories; panel (b) compares the resulting long-run equilibrium toxicity levels under the three policy scenarios. The figure matters because it shows both the empirical support for the RE proxy and the remaining identification gap in the pressure block.

7.7 External Jigsaw text toxicity reference

The Jigsaw toxicity dataset [12], when supplied separately by the user in `data/toxicity/train.csv`, provides an external text toxicity reference for the magnitude of τ. It is not part of the core Higgs dataset and is not required for the main spread calibration. This dataset contains text level toxicity labels from a different platform context, namely Wikipedia talk pages. We use it only to calibrate the *scale* of τ, not to fit the toxicity dynamics. The mean Jigsaw toxicity rate is approximately 0.1017, which gives an external scale comparison value of $\phi \approx 0.1459$ when $\psi = 0.1$ and the moderate scenario equilibrium viral volume is used for scaling.

7.8 Reference-constrained calibration

The pressure parameters (ϕ, ψ) are not uniquely identified from the same-dataset proxy alone. The unconstrained fit achieves good proxy agreement ($R^2 \approx 0.74$) but at a pressure scale ($\tau^*_{\text{mod}} \approx 0.067$) that is inconsistent with the external Jigsaw reference. We resolve this identification problem partially through a *reference-constrained* approach.

The external Jigsaw text toxicity rate anchors the steady state gain ratio:

$$r = \frac{\phi}{\psi} = \frac{\tau_{\text{ref}}}{V^*_{\text{mod}}} \approx \frac{0.1017}{0.0697} \approx 1.46.$$

This constrains (ϕ, ψ) to the line $\phi = r\psi$ in parameter space. We then treat ψ as the single free parameter on this constraint line and choose the value that best matches the same-dataset RE proxy dynamics. This one-dimensional optimization gives:

$$\phi_{\text{rc}} \approx 0.417, \qquad \psi_{\text{rc}} \approx 0.286,$$

with proxy-fit $R^2 \approx 0.41$.

Table 4 compares all pressure-block configurations.

Table 4: Pressure-block configurations compared against the selected RE proxy.

Configuration	ϕ	ψ	Proxy R^2	Moderate τ^*
Current baseline	0.0560	0.1000	0.1844	0.0390
Higgs decay-anchored	0.0613	0.0409	0.0952	0.1043
External scale	0.1459	0.1000	0.3091	0.1015
Reference-constrained	0.4174	0.2860	0.4064	0.1015
Higgs (boundary-constrained)[1]	0.9596	1.0000	0.7422	0.0668

The reference-constrained configuration achieves better proxy-fit quality ($R^2 \approx 0.41$) than either the baseline ($R^2 \approx 0.18$) or the plain external-scale setting ($R^2 \approx 0.31$), while maintaining the externally anchored pressure scale. The remaining gap between the constrained ($R^2 \approx 0.41$) and boundary-constrained ($R^2 \approx 0.74$) fits is the substantive identification limit of the one-timescale pressure block. Closing this gap would require either a dataset with both spread data and directly observed toxicity labels, or a multi-timescale τ model.

The empirical picture the calibration establishes is: the spread side is tightly identified (bootstrap 95% CI on α_{crit} spans [0.313, 0.320]), but the toxicity block is constrained only along the ratio $\phi/\psi \approx 1.46$. The individual values of ϕ and ψ require an independent text toxicity measurement, or an independent same platform toxicity measurement, to pin down separately.

We also checked whether the spread-side conclusions are sensitive to the exact calibration window and time discretization. Re-fitting the reduced IVF model on 80, 100, and 120 hour windows with the original 1 hour discretization leaves the main qualitative conclusions intact: the dominant spike is still well fit and the inferred spread parameters remain in the same general range. A harsher stress test with 2 hour binning performs much worse, which is informative rather than alarming. The model was formulated and calibrated at the 1 hour event scale, so coarse aggregation compresses the spike shape and weakens fit quality. This matters because it shows the main spread conclusion is robust to nearby window choices at the native time scale, while also clarifying the temporal resolution at which the calibration should be interpreted.

8 Numerical results

8.1 Policy scenarios

We compare three amplification scenarios that represent different platform policies:

- **Engagement-first** ($\alpha = 0.9$): aggressive amplification, representing a platform that prioritizes short-run engagement metrics.
- **Moderate** ($\alpha = 0.5$): balanced amplification, representing a platform that partially accounts for content quality alongside engagement.
- **Health-first** ($\alpha = 0.2$): conservative amplification, representing a platform that prioritizes user well-being over engagement metrics.

Table 5 reports the long-run equilibrium values under each scenario with the baseline pressure parameters.

[1] The optimizer reached the upper bound $\psi = 1.0$ imposed during optimization; the true unconstrained optimum may have $\psi > 1$.

Table 5: Long-run equilibrium values under the three policy scenarios (baseline pressure block).

Scenario	$\mathcal{R}_0$	V^*	τ^*	U^*	Regime
Engagement-first ($\alpha = 0.9$)	2.84	0.1203	0.0674	29.88	Supercritical
Moderate ($\alpha = 0.5$)	1.58	0.0697	0.0390	35.97	Supercritical
Health-first ($\alpha = 0.2$)	0.63	0.0000	0.0000	50.00	Subcritical

The results show a clear pattern:

1. **The health-first scenario eliminates sustained virality.** With $\mathcal{R}_0 = 0.63 < 1$, viral content cannot sustain itself, and the system converges to the DFE with $V^* = \tau^* = 0$ and $U^* = \nu/\lambda_u = 50$. This is the maximum possible user retention.

2. **Higher amplification increases viral volume and pressure.** Moving from moderate to engagement-first increases V^* by 73% (from 0.070 to 0.120) and τ^* by 73% (from 0.039 to 0.067). The proportional increase is the same because $\tau^* = \phi V^*/\psi$ is linear in V^* at equilibrium.

3. **Higher amplification reduces user retention.** The engagement-first scenario retains only 29.9 users compared to 36.0 in the moderate scenario and 50.0 in the health-first scenario. This is a 40% reduction relative to the health-first baseline ($U^* = 50$).

4. **The tradeoff is real but nonlinear.** The engagement-first scenario does produce more short-run viral content, but the long-run user base is substantially smaller. Whether this tradeoff is favourable depends on what the platform optimizes.

8.2 Continuation in amplification

Figure 9 shows the results of a continuation sweep over $\alpha \in [0.05, 1.0]$. The sweep reveals:

- A clear threshold crossing at $\alpha \approx 0.3165$. Below this point, $V^* = \tau^* = 0$.
- Above the threshold, both V^* and τ^* increase monotonically with α.
- U^* decreases monotonically with α above the threshold, reflecting pressure-dependent attrition.
- The engagement measure $E^* = \alpha V^* U^*$ increases with α in the tested range, meaning that higher amplification does produce more total engagement, but at the cost of a smaller, more pressured user base.

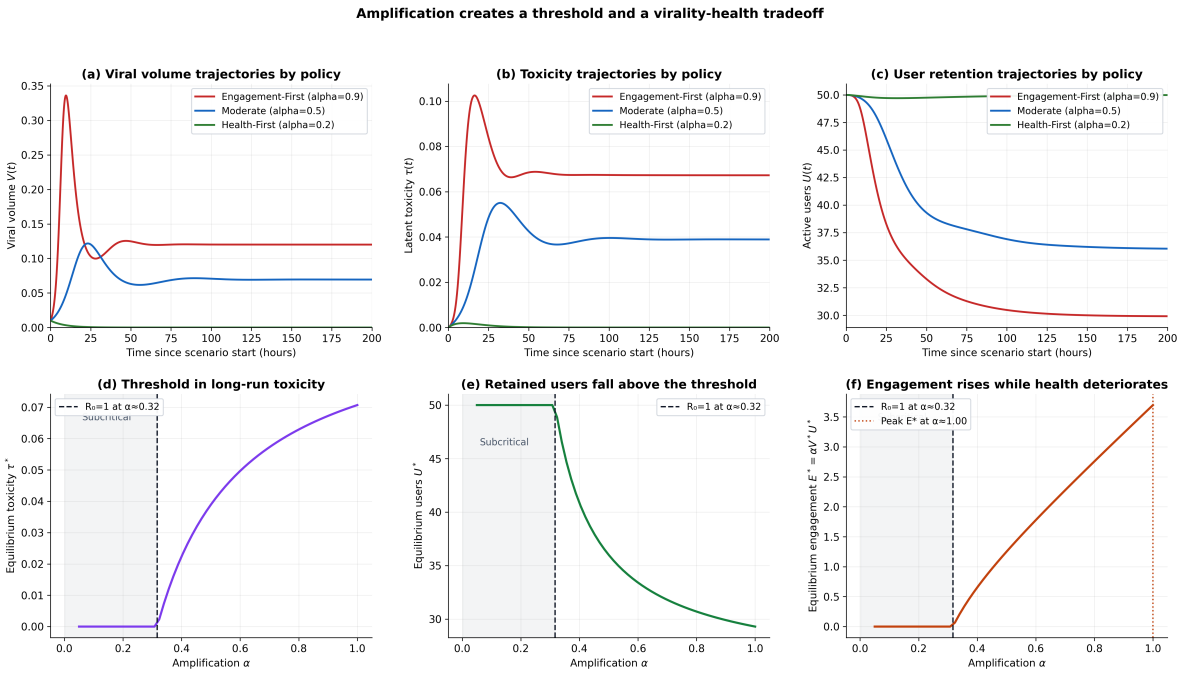

Figure 9: Main policy evidence from the IVFS model. The top row shows time trajectories for viral volume, latent toxicity, and retained users under the health-first, moderate, and engagement-first scenarios; the bottom row shows how equilibrium toxicity, retained users, and engagement change as amplification α varies continuously. The dashed threshold line at $\alpha_{\text{crit}} \approx 0.3165$ and the shaded subcritical region make the central tradeoff visually obvious: stronger amplification raises sustained virality and toxicity while reducing the long-run user base.

8.3 Pressure sensitivity

We perform a local sensitivity sweep over the pressure forcing parameter ϕ with ψ fixed at its baseline value of 0.1. The purpose is to check whether the qualitative policy ranking (engagement-first > moderate > health-first in terms of pressure) is robust to changes in ϕ.

Figure 10 shows the results. Across the entire tested range of ϕ, the ordering of the three scenarios is preserved: the engagement-first scenario always produces the highest long-run pressure, the moderate scenario remains in the middle, and the health-first scenario stays at zero pressure because it is subcritical. The baseline value $\phi = 0.056$ and the external-reference value $\phi \approx 0.146$ are both within the range where the qualitative conclusions hold.

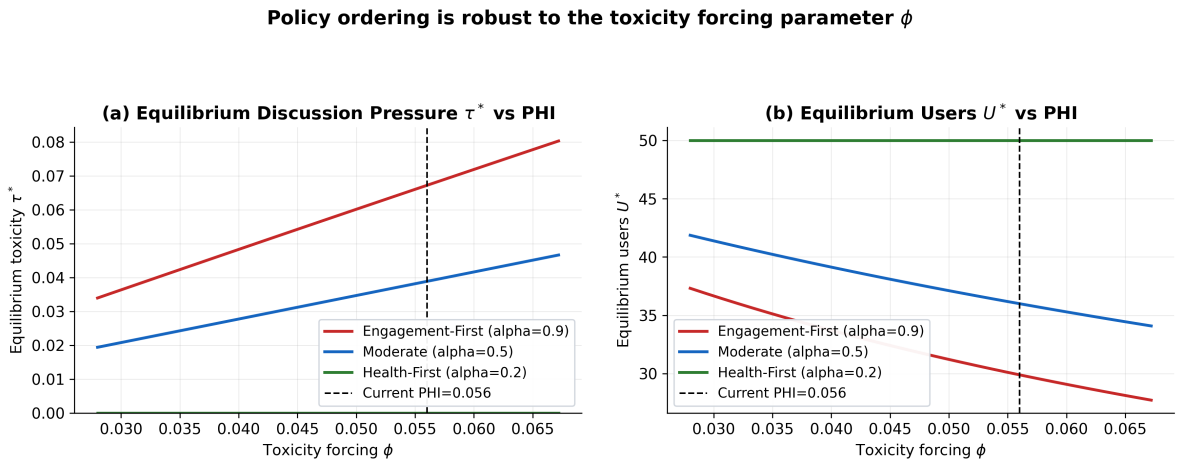

Figure 10: Robustness of the policy ordering to the toxicity forcing parameter ϕ with $\psi = 0.1$ fixed. The left panel tracks equilibrium toxicity and the right panel tracks equilibrium users across a range of ϕ values, with vertical markers for the baseline and external-reference scales. The qualitative ranking of the three policies does not change, which matters because the pressure block is only partially identified and the main policy conclusion should not depend on one single choice of ϕ.

Because the pressure parameters are only partially identified, it matters that the qualitative policy ranking holds across a wide range of ϕ values. This gives the main conclusions more stability than the individual parameter estimates alone would suggest.

8.4 Multi-window validation

Because the Higgs record spans only 168 hours, independent 100-hour windows are not available. To assess whether the spread model generalizes beyond the main calibration window, we refit the reduced IVF helper model on three representative 48-hour active windows from the same record. Note that these windows overlap substantially (e.g., windows 2 and 3 share hours 120 to 151, a 32-hour overlap), so the refits are not independent; the purpose is to test whether the model shape fits different phases of the cascade, not to provide independent replication.

Table 6: Active-window validation for the reduced IVF helper model.

Window (hours)	Peak RT count	Local fit R^2	Interpretation
76 to 123	30,602	0.9514	Main spike local refit
104 to 151	3,033	0.7170	Secondary active tail
120 to 167	1,498	0.6926	Late active tail

The local R^2 values range from 0.69 to 0.95, which shows that the burst-and-decay shape captured by the IVF helper model is a consistent feature of the cascade across different phases. The fitted parameters do shift across windows, which is expected: different phases of the cascade have different intensity and duration. The correct interpretation is local phenomenological robustness: the model shape works across multiple active phases, rather than globally constant parameters.

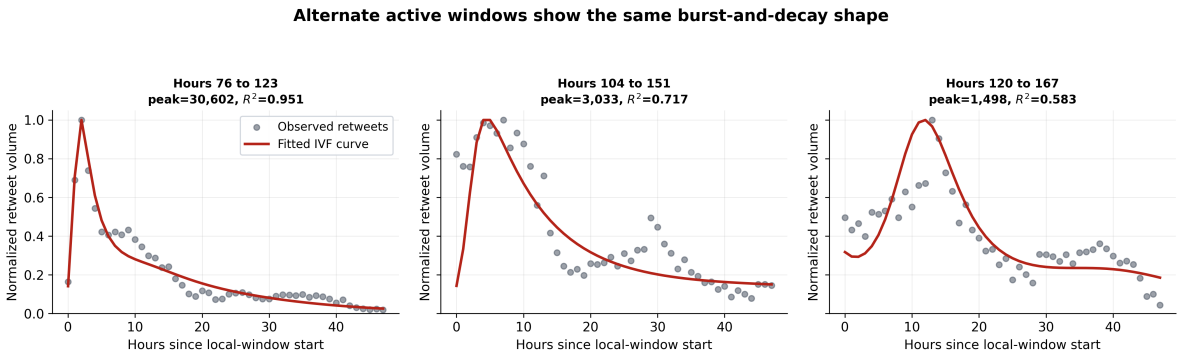

Figure 11: Representative 48-hour active-window refits drawn from different phases of the Higgs cascade. Each panel shows observed retweet counts and the locally fitted IVF curve for one active window. The takeaway is not that the parameters are identical across windows, they are not, but that the same burst-and-decay mechanism continues to fit the event at its native one-hour scale, which strengthens the empirical case that the main calibration is not a one-window accident.

The policy story emerging from these results is consistent across all tested pressure calibrations: any platform operating above α_{crit} faces a structural tradeoff in which amplification gains in viral engagement are paired with rising toxicity and accelerating user departure. The ordering, higher α means higher V^*, higher τ^*, and lower U^*, is not an artefact of a particular parameter choice; it follows from the model's feedback structure.

9 Limitations

Identification limitations. The pressure parameters (ϕ, ψ) are not jointly identified from the available data. The spread block is well-identified (bootstrap CIs are narrow, profile likelihood shows clear minima), but the pressure block relies on a reply proxy that does not separately pin down the forcing and relaxation timescales. The reference-constrained calibration resolves this partially: it anchors ϕ/ψ to the Jigsaw external reference but cannot determine the individual values. The boundary-constrained fit (Table 4) reaches the upper bound $\psi = 1.0$ imposed during optimization; the true optimum may lie at $\psi > 1$, and relaxing this bound is a natural next step once independent pressure measurements are available.

Data limitations. The spread model is fitted on a single event cascade (Higgs boson announcement, July 2012). The multi-window validation (Section 8.4) confirms that the IVF burst-and-decay shape generalises across phases of this event, but the parameter values are specific to this cascade and this platform. Calibration on independent events from different topics, time periods, and platforms is needed to assess whether the threshold value and spread rates carry over.

Modelling limitations. Three structural simplifications bound the model's scope. First, homogeneous mixing: all content items in a compartment interact identically, ignoring network topology, community structure, and content heterogeneity. Second, deterministic dynamics: the model cannot capture stochastic fluctuations or rare-event behaviour inherent in individual sharing decisions. Third, scalar pressure: the latent state τ collapses the multidimensional nature of online toxicity (hate speech, harassment, misinformation) into a single index; a richer model might track multiple pressure types separately.

External validity limitations. The rage-bait operational definition (high τ/V ratio, Section 1.5) is validated here only against the structural signature present in a scientific announcement cascade, not against ground-truth toxicity annotations. Future work should test whether the structural signature coincides with content independently labelled as inflammatory (e.g. using Jigsaw or Ruddit labels). Similarly, the DFE-based $\mathcal{R}_0$ derivation uses linearization of the $\dot{V}$ equation; a full next-generation matrix treatment would confirm the result and clarify the roles of individual model components.

Despite these limitations, the model robustly supports the existence of a sharp amplification threshold and a monotone amplification retention tradeoff. What it cannot yet establish is whether the specific threshold value $\alpha_{\text{crit}} \approx 0.3165$ or the magnitude of the retention loss generalise across platforms, content types, or event scales.

10 Contributions of each group member

Neel Sarkar was responsible for the mathematical formulation of the IVFS model, the implementation of all calibration, simulation, and analysis code, the equilibrium and stability analysis, the sensitivity computations, the benchmark comparisons, and the writing of this report.

Piper Scott contributed to the initial problem formulation and motivation, assisted with literature review, helped select and interpret the Higgs dataset, and reviewed drafts of the report.

11 Conclusion

11.1 Summary of results

We have developed and analyzed a deterministic compartmental ODE model for content lifecycle dynamics on social media. The model tracks six state variables: four content lifecycle compartments (ignored, viral, fatigued, suppressed), a latent discussion-pressure state, and an active-user pool. The key results are:

1. **Well-posedness.** The model preserves nonnegativity and admits a bounded invariant region (Propositions 1 and 2).

2. **Clear amplification threshold.** Viral content is self-sustaining if and only if $\mathcal{R}_0 > 1$, which occurs at $\alpha > \alpha_{\text{crit}} \approx 0.3165$ under the current calibration. The transition is consistent with a transcritical bifurcation.

3. **Strong spread calibration.** The reduced IVF helper model fits the Higgs retweet data with full-window $R^2 = 0.92$, and the spread parameters are tightly identified (bootstrap CI width < 0.01). The IVF model outperforms a classical SIR benchmark on the spike window (spike-window $R^2 = 0.89$ vs. 0.69).

4. **Robust policy ranking.** Across all tested pressure configurations, the qualitative ordering is preserved: higher amplification increases viral volume and discussion pressure while lowering user retention.

5. **Pressure identification.** The pressure block is partially identified through a combination of same-dataset reply proxy and external toxicity reference. The reference-constrained calibration provides the best available estimate ($R^2 \approx 0.41$), but a gap remains relative to the boundary-constrained optimum ($R^2 \approx 0.74$, where the optimizer reached the upper bound on ψ).

11.2 Significance

The central result is that a sharp amplification threshold ($\alpha_{\text{crit}} \approx 0.3165$) governs whether virality and discussion pressure self-sustain. Below it, both collapse to zero and the platform retains its full user base. Above it, the system locks into a feedback cycle whose equilibrium viral volume, pressure level, and user attrition all increase monotonically with how aggressively the platform amplifies.

The distinctive feature of this result is that it follows from a *content centric* model. Prior compartmental work on social media dynamics almost uniformly classifies users into behavioural states, susceptible, infected, recovered, and variants thereof. This requires knowledge of individual user exposure histories that are rarely available in public data, and it places the policy variable (amplification) as an implicit modifier of user to user transmission rates. By instead classifying content items into lifecycle stages, the IVFS state variables correspond directly to quantities platforms observe and control: content volume in each visibility tier, reply rate as a toxicity signal, and active user count. The amplification parameter α is a platform control variable, not a latent individual trait.

This alignment between model state and platform observables makes the threshold analysis directly actionable. Determining whether α clears α_{crit} requires only content level event counts, not inferred user state data. A six state ODE calibrated on a single real world cascade is sufficient to produce a quantitative threshold and a policy scenario comparison with interpretable equilibrium values, a finding that would not emerge from a user centric model of the same cascade.

11.3 Next steps

Some directions for future work:

1. **Next-generation matrix for $\mathcal{R}_0$.** The current threshold is derived from DFE linearization of the $\dot{V}$ equation. A full next-generation matrix derivation would provide a more rigorous foundation and clarify the roles of the different model components in the reproduction number.

2. **Global stability.** We have established local stability of both the DFE (Proposition 3) and the positive equilibrium (numerically). Global stability results (for example, using Lyapunov functions) would strengthen the analytical foundation.

3. **Bifurcation analysis.** The transcritical bifurcation at $\mathcal{R}_0 = 1$ could be studied more formally using center manifold reduction. Additionally, we have not ruled out the possibility of Hopf bifurcations at extreme parameter values, which would produce oscillatory dynamics.

4. **Multi-cascade validation.** Calibrating the model on multiple independent event cascades from different platforms and topics would test the generalizability of the content-centric framework.

5. **Multi-timescale pressure.** Replacing the one-timescale pressure block with a two-timescale or state-dependent model could resolve the identification gap between the constrained and unconstrained pressure fits.

6. **Stochastic extension.** A stochastic differential equation version of the model would allow us to study the effects of noise on threshold behaviour and equilibrium fluctuations.

7. **Network-aware extension.** Incorporating network structure (e.g., through degree-dependent transmission rates) would relax the homogeneous mixing assumption and allow the model to capture community-level dynamics.

A Reproducibility notes

All code used in this project is available in the repository. The main scripts are:

```
python -m code.ivfs_validation --offline
python -m code.equilibrium_analysis --offline
python -m code.benchmark_models --offline
python -m code.toxicity_calibration
```

The main outputs (figures, diagnostics, and run metadata) are written to the `assets/` directory. The Python environment is specified in `requirements.txt`. The Higgs dataset is expected locally, with optional external toxicity files supplied separately by the user.

B Identification note for the pressure block

The empirical package identifies different parts of the model with different strengths:

- **Higgs retweet counts** strongly identify the spread-side parameters $(\beta_0, \gamma_0, \lambda_0, d_\lambda, V_0)$ on the main 100-hour window. Bootstrap CIs are narrow and profile likelihood curves show clear minima.
- **Higgs reply activity** partially identifies the pressure *shape*: it distinguishes better and worse relaxation timescales ψ, but only through a proxy, and therefore without fixing an absolute pressure scale.
- **The Jigsaw reference** identifies the steady-state gain ratio $r = \phi/\psi = \tau_{\mathrm{ref}}/V^*_{\mathrm{mod}}$, constraining (ϕ, ψ) to a line rather than to a point.
- **The reference-constrained fit** combines items two and three: it restricts (ϕ, ψ) to the externally anchored line $\phi = r\psi$ and then chooses the ψ on that line that best matches the RE proxy. This is the most informative single estimate currently available, but it leaves a residual gap relative to the unconstrained same-dataset optimum.

C Possible model extensions

Natural extensions of the present framework include:

1. A full next-generation matrix derivation of $\mathcal{R}_0$.
2. Theorem-level local and global stability results using Lyapunov methods.
3. A multi-timescale or state-dependent pressure model.
4. A latent observation model for τ with RE and MT as noisy observables.
5. Independent-event spread validation beyond the Higgs record.
6. Network-aware formulations with degree-dependent rates.
7. Stochastic extensions for uncertainty quantification in the dynamics.

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